

User Manual

The ASL-DVS Suite

How to use the ASL-DVS web tools day to day: what every page, tab and button does, with a technical close to each chapter for when you want to know how it works. Written for the Pi Zero 2 W build; full-build differences are marked where they matter.

[DASHBOARD](#)[SYSMON](#)[INSTMON](#)[WIFIMON](#)[44HELPER](#)

FILES COVERED BY THIS EDITION

TOOL	FILE	VERSION
Dashboard	<code>asl_dvs_dashboard_pi02w_v9_3_71_13_20261007.py</code>	9.3.71.13-pi02w
SysMon	<code>sysmon_pi02w_v6_13_67_9_20261007.py</code>	6.13.67.9-pi02w
44helper	<code>asl_dvs_m17_44helper_v0_0_175_20261008.py</code>	0.0.175
instmon	<code>instmon_v2_38_3_20261007.py</code>	2.38.3
wifimon	<code>wifimon_v5_28_20261007.py</code>	5.28
Installer	<code>install_asl_dvs_v6_6_20261006.sh</code>	6.6
SysMon FULL	<code>sysmon_v6_13_74_20261007.py</code>	6.13.74
Dashboard FULL	<code>asl_dvs_dashboard_v9_3_73_20261006.py</code>	9.3.73

[DRAFT · PART I OF V](#)

In this manual

PART I · GETTING STARTED

- | | | |
|----------|-----------------------------------|------------|
| 1 | The suite at a glance | THIS DRAFT |
| 2 | Installing and updating, in brief | THIS DRAFT |

PART II · SYSMON

- | | | |
|----------|---|-------------|
| 3 | SysMon basics and live status: Overview, Services, Ports, Journal | LATER STAGE |
| 4 | Node and DVSwitch health: ASL-DVS, DVSM, STFU, M17, Phone | LATER STAGE |
| 5 | System care: Tune, Hardware, SD Card, Security, Edit | LATER STAGE |
| 6 | Full-build tabs: Firewall, Reg, Net, Zello, Console | LATER STAGE |

PART III · 44HELPER

- | | | |
|-----------|---|-------------|
| 7 | 44helper basics: Overview, Services, Ports, Firewall, Actions Log | LATER STAGE |
| 8 | 44Net and remote access: Router Install, Tunnels, Nodes, Cloudflare | LATER STAGE |
| 9 | Software installers: ASL3, DVSwitch, SVXLink, M17, Ampersand | LATER STAGE |
| 10 | Keeping the system healthy: Update, System Optimization | LATER STAGE |
| 11 | The Config tab: restore points, save points, presets, identity | LATER STAGE |

PART IV · THE OTHER TOOLS

- | | | |
|-----------|--------------------------|-------------|
| 12 | The Dashboard | LATER STAGE |
| 13 | instmon | LATER STAGE |
| 14 | wifimon and wifi_menu.sh | LATER STAGE |

PART V · REFERENCE

- | | | |
|-----------|--|-------------|
| 15 | Troubleshooting and recovery | LATER STAGE |
| A | Appendices: ports, files, services, launcher, glossary | LATER STAGE |

How to use this manual

If you are new to the suite, read Part I first, then turn to the chapter for the tool in front of you. Every chapter after Chapter 2 follows the same order, so you always know where to look:

- 1 What it's for:** what the tool or tab does, and when you would open it.
- 2 Before you start:** what has to be installed, and anything that can go wrong.
- 3 Tab by tab:** each card and button, and what good and bad readings look like.
- 4 Common tasks:** short recipes, step by step.
- 5 Under the hood:** the technical close, in a purple-labelled panel. It covers the files, services, ports and timers involved. You can skip it on a first read.

Markers used in the text

NOTE Background, or something worth knowing.

TIP A quicker or safer way to do something.

CAREFUL This changes your node, restarts something or needs a reboot.

FULL BUILD Applies to the full build (Pi 3, 4 or 5) only. Also shown as the badge **FULL**.

Commands and pictures

Commands are typed on the Pi, in an SSH session or at a keyboard and screen. Lines starting with **#** are notes. Replace `<pi-address>` with your Pi's IP address or host name.

```
# a note about the next line
sudo systemctl status sysmon
```

Pictures marked **Illustration** are drawn to show a page's layout. They use the same tab names and order as the real page, but the values in them are made up. Example identities use the placeholders `NØCALL` and `<NODE>`.

PART I

Getting started

What the suite is, what each tool is for, how to log in, and the short version of installing and updating it. The full install steps are in the install manuals in the [docs/](#) folder.

1. The suite at a glance
2. Installing and updating, in brief

The suite at a glance

ASL-DVS is a set of five small web tools that run on the same Raspberry Pi as your AllStarLink node. Each tool has its own page in the browser. Together they let you run, check, repair and update the node without typing commands.

1.1 What is on the node

The tools sit on top of a working AllStarLink 3 (ASL3) node with DVSwitch. It helps to know the main parts, because the tools check and talk to all of them:

PART	WHAT IT DOES
Asterisk + app_rpt	The ASL3 engine. It runs your node, links to other AllStarLink nodes over IAX2 and handles EchoLink.
Analog_Bridge	Turns the node's audio into digital voice and back again. It talks to Asterisk on a private node (1999 by convention).
MMDVM_Bridge	Connects Analog_Bridge to the digital networks: DMR directly, and System Fusion, P25, NXDN and D-STAR through their gateways.
Gateways	YSFGateway, P25Gateway, NXDNGateway and ircDDBGateway. Each one connects to that mode's reflectors or talkgroups.
USRP2M17	Bridges a second private node (1917 by convention) to an M17 reflector.

```
AllStarLink network
| IAX2 (UDP 4569)
[ ASL3 / Asterisk ]----- your node <NODE>      (radio, optional)
| USRP (UDP, this Pi only)
+-- private node 1999 -- Analog_Bridge -- MMDVM_Bridge -- DMR network
|                                     |
|                                     YSFGateway / P25Gateway / NXDNGateway / ircDDBGateway
|
+-- private node 1917 -- USRP2M17 ----- M17 reflector
```

Figure 1.1 The audio path through a bridge node. Node numbers 1000 to 1999 are private: they never leave the Pi. The full layout is in [docs/ASL-DVS_Idealized_Node_Configuration.pdf](#).

1.2 The five tools

Each tool is one file, runs as its own service, and serves its own page on its own port.

TOOL	PORT	WHAT IT'S FOR	OPEN IT WHEN YOU WANT TO...
Dashboard	8989	Node control.	Connect to AllStar or EchoLink nodes, change DMR talkgroup or YSF, P25, NXDN, D-STAR, XLX or M17 reflector, use favorites, make a phone call.
SysMon	9999	System monitor and health checks.	See whether the Pi and node are healthy, find a stopped service or a bad config file, read logs, look after the SD card.
44helper	9997	Setup, repair and configuration helper.	Set up 44Net or Cloudflare remote access, install ASL3, DVSwitch, SVXLink, M17 or Ampersand, run updates, save and restore your config.

instmon	8990	Installer and manager for the suite.	Update the tools from GitHub, start or stop them, pause the suite for a big job, back up or image the SD card.
wifimon	8991	WiFi and power watchdog.	Change WiFi networks, check signal and supply voltage, set up the automatic safe shutdown.

NOTE 44helper is optional. The other four make up the core suite. A sixth item, the Watchdog script, is no longer needed: the Dashboard's own service restarts it if it stops.

Which tool for which job

I WANT TO...	GO TO
Talk on a DMR talkgroup or a reflector	Dashboard, the mode's tab
Link to another AllStar node	Dashboard, ASL tab
Find out why something stopped working	SysMon, Overview and Services, then the matching check tab
Check that my config files agree with each other	SysMon, ASL-DVS and DVSM tabs; 44helper, Config tab
Save my config before I change it	44helper, Config tab (restore point)
Reach my node from outside my home network	44helper, Router Install / Tunnel tabs or Cloudflare
Update the suite	instmon, GitHub Updates
Update Linux and ASL packages	44helper, Update tab
Make the SD card last longer	SysMon, SD Card tab; 44helper, System Optimization
Change WiFi network	wifimon

1.3 A first look at each page

Every page has a header with the tool's name and version and a Log Out button. Most pages have a row of tabs under the header; click a tab to switch view. The pictures below show each tool's tabs in order.

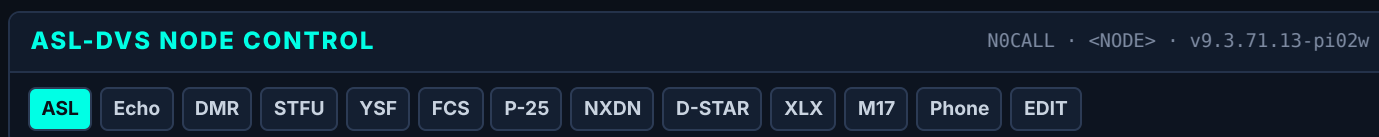


Figure 1.2 · Illustration. The Dashboard (port 8989). One tab per mode, plus Phone and EDIT. EDIT holds your favorites and settings. The Pi Zero 2 W build keeps every tab of the full build.

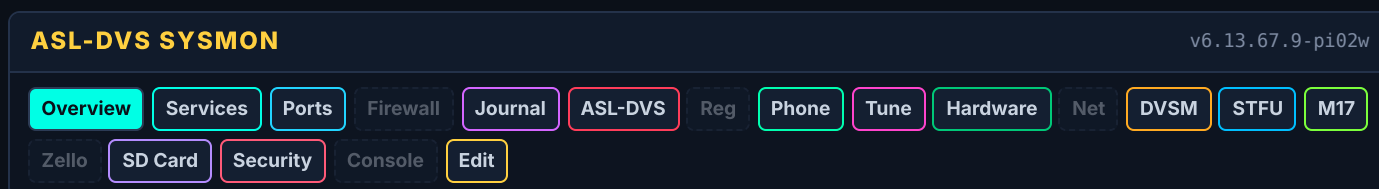


Figure 1.3 · Illustration. SysMon (port 9999). The 14 solid tabs are in the Pi Zero 2 W build. The 5 faded tabs (Firewall, Reg, Net, Zello, Console) are **FULL** build only.

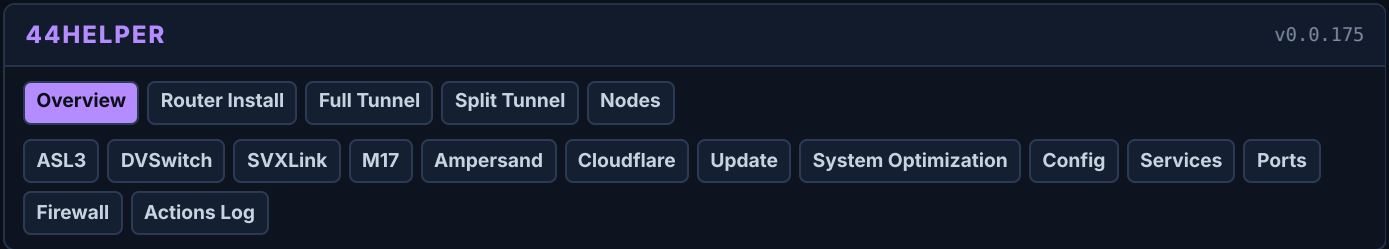


Figure 1.4 · Illustration. 44helper (port 9997). The first row is 44Net remote access. The second row has the installers, maintenance, Config and the status tabs. 44helper is the same on every Pi.

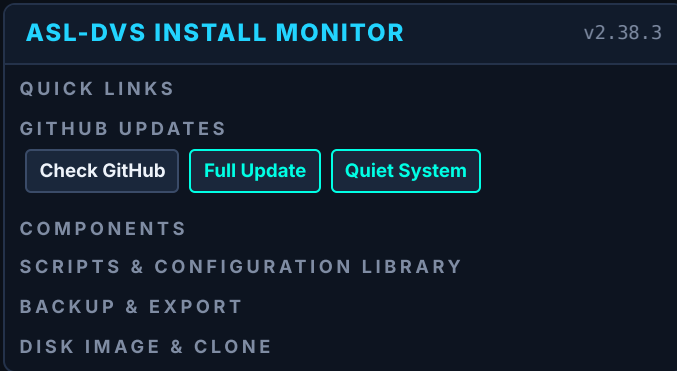


Figure 1.5 · Illustration. instmon (port 8990) is one long page of sections, not tabs.

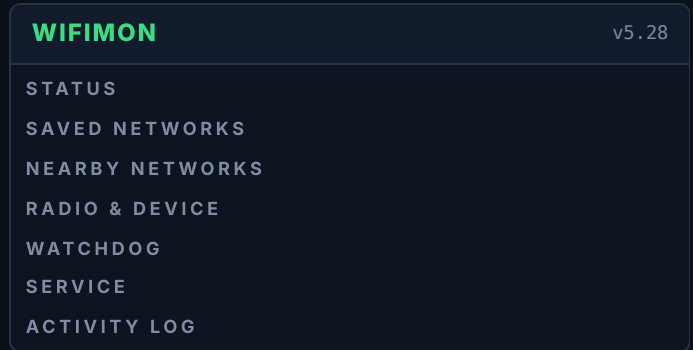


Figure 1.6 · Illustration. wifimon (port 8991) is also one page of cards.

1.4 Two builds: Pi Zero 2 W and full

The Dashboard and SysMon come in two builds. The other tools have a single build that runs on every Pi.

PI ZERO 2 W BUILD **PI02W**

For the 512 MB Pi Zero 2 W. This manual is written for it.

- Dashboard: every tab is kept. It searches the AllStar and EchoLink lists from disk instead of memory, and refreshes less often.
- SysMon: drops the Firewall, Reg, Net, Zello and Console tabs to save memory.

FULL BUILD **FULL**

For a Pi 3, 4 or 5.

- Dashboard: the same tabs, and it keeps the lists in memory.
- SysMon: all 19 tabs. Chapter 6 covers the five extra ones.

Which build am I running? Look at the version in the page header. A Pi Zero 2 W build version ends in **-pi02w**, for example **6.13.67.9-pi02w**. A full build version doesn't, for example **6.13.74**.

NOTE Apart from SysMon, the builds look and work the same, so this manual points out the full build only where it differs.

1.5 Logging in

- 1 Open the page in a browser: **http://<pi-address>:<port>**, for example **http://192.168.1.50:9999** for SysMon.
- 2 Type the Pi's **root password** and press Log In. There is no user name.
- 3 When you're done, press **Log Out** in the header, especially on a shared computer.

- **Each tool has its own login.** Logging in to SysMon doesn't log you in to the Dashboard.
- **Wrong passwords lock you out for a while.** After several wrong tries in a row, that address has to wait before it can try again, and the wait grows with each failure. Wait it out; don't keep guessing.
- **44helper stays logged in** until you log out, press **Log out all devices**, or the Pi reboots, because its jobs can run for a long time.

CAREFUL The pages use plain HTTP, so your password crosses the network unencrypted. Only open them on your own network or over a VPN or tunnel, never straight from the internet. Chapters 7 and 8 cover safe remote access with 44helper.

TIP If the login is always refused, set the root password again on the Pi with `sudo passwd root`. Raspberry Pi OS ships with root locked, so a new Pi needs this once.

1.6 Phones and tablets

The pages are laid out for phone, tablet and desktop widths. On a phone, wide tables hide their less important columns, and the tab row wraps onto more lines. Turning a tablet to landscape shows everything.

Leave a page open only while you're using it. Each open page polls its tool for fresh data, and on a Pi Zero 2 W that costs CPU and memory.

UNDER THE HOOD

Services, files and ports

TOOL	SYSTEMD SERVICE	PROGRAM	CONFIG
Dashboard	asl_dvs_dashboard	/usr/local/bin/asl_dvs_dashboard.py	/etc/asl_dvs/asl_dvs.conf
SysMon	sysmon	/usr/local/bin/sysmon.py	/etc/sysmon/sysmon.conf
44helper	44helper	/opt/44helper/asl_dvs_m17_44helper.py	/etc/44helper/44helper.conf
instmon	instmon	/usr/local/bin/instmon.py	/etc/asl_dvs/instmon_library/
wifimon	wifimon	/usr/local/bin/wifimon.py	/etc/wifimon/wifimon.conf

The Dashboard and SysMon program paths are links into `/usr/local/lib/asl_dvs/` and `/usr/local/lib/asl_dvs_sysmon/`. The installed file keeps its own name, so the version is always visible on disk.

One process each, started through the launcher

Every service starts its tool through the shared launcher `/usr/local/bin/asl_dvs_launch.py`:

```
ExecStart=/usr/bin/python3 /usr/local/bin/asl_dvs_launch.py /usr/local/bin/sysmon.py
```

The launcher loads the tool as a module so that Python keeps its compiled copy in `__pycache__`. That roughly halves each tool's memory (SysMon on a Pi Zero 2 W: 54 MB started directly, 26 MB through the launcher). The launcher also turns on three memory savers: `python3 -O0`, `MALLOC_ARENA_MAX=2` and a periodic `malloc_trim`. Each has an off-switch file in `/etc/asl_dvs/` named `launch_no_optimize`, `launch_no_arena_cap` or `launch_no_trim`.

How a page works

- Each tool is a small Python HTTP server. The page's HTML, CSS and JavaScript are built into the tool's file, so there is nothing else to install and no web server to configure.
- The page fetches live data from the tool's own `/api/...` addresses as JSON and redraws its cards.
- The login checks the password against the root entry in `/etc/shadow`, and sets a session cookie. Requests that change anything also need a matching CSRF header, so another web page can't act for you.
- Version numbers come from the `VERSION` line in each file. The `-pi02w` suffix is how the installer and instmon recognise the Pi Zero 2 W build.

Checking on a tool from the shell

```
# is it running, and its last 30 log lines
systemctl status sysmon --no-pager
journalctl -u sysmon -n 30 --no-pager
# is anything listening on its port?
ss -tlnp | grep 9999
# restart all the web tools
sudo systemctl restart sysmon asl_dvs_dashboard instmon wifimon 44helper
```

Installing and updating, in brief

This chapter covers just enough to get the suite onto a node and keep it current. For every step and option, use the install manual for your build.

MANUAL	FILE IN DOCS/
Pi Zero 2 W install	ASL-DVS_Pi02w_Install_Manual.pdf
Full build install FULL	ASL-DVS_Full_Build_Install_Manual.pdf

2.1 Before you start

- A Raspberry Pi already running AllStarLink 3 with DVSwitch set up.
- SSH access with `sudo`, and a root password (`sudo passwd root`).
- On a Pi Zero 2 W, some free memory. Check with `free -m`. The install manual lists services you can turn off if you don't use them.

2.2 Install in three steps

- 1 Get the files.** Download the current files from github.com/kd8pgk/ASL-DVS into one folder on the Pi.
- 2 Install instmon first.** It can then install and update everything else from the browser.

```
sudo python3 instmon_v2_38_3_20261007.py --install
```

- 3 Install the rest** with the installer. On a new node it asks for your callsign, node number and a label.

```
sudo bash install_asl_dvs_v6_6_20261006.sh
```

The installer picks the build for you: the Pi Zero 2 W build on a Pi Zero 2 W, the full build on anything else. If one build is already installed, it keeps that build. When it finishes, it checks that each tool is running and listening on its port.

TIP 44helper is optional. Install it the same way, or from instmon's page, when you want its remote-access, installer or Config tools.

2.3 First visit to each page

PAGE	ADDRESS	CHECK FIRST
Dashboard	<a href="http://<pi-address>:8989">http://<pi-address>:8989	Your callsign and node number are right in the header.
SysMon	<a href="http://<pi-address>:9999">http://<pi-address>:9999	The Overview shows no red items (Chapter 3).
instmon	<a href="http://<pi-address>:8990">http://<pi-address>:8990	Every tool shows as installed and running.
wifimon	<a href="http://<pi-address>:8991">http://<pi-address>:8991	Answer "Turn the watchdogs on?" (below).
44helper	<a href="http://<pi-address>:9997">http://<pi-address>:9997	Take a restore point on the Config tab (Chapter 11).

wifimon's watchdogs

- A new wifimon starts with both watchdogs off: it shows WiFi and voltage, but won't act on them. On your first login it asks whether to turn them on. **Add your WiFi networks first.**
- **WiFi watchdog:** reconnects to your saved networks, and shuts the Pi down after 3 minutes with no connection.
 - **Voltage watchdog:** shuts the Pi down after 30 seconds of low supply voltage.

CAREFUL Both watchdogs can shut the node down. A Pi Zero 2 W is sensitive to voltage drop, so use a good power supply and a short cable before turning the voltage watchdog on.

2.4 Keeping the suite up to date

Updates come from GitHub through instmon's **GitHub Updates** section.

GITHUB UPDATES

Check GitHub

Full Update

3 newer on GitHub · checked 14:02

Pi Zero 2 W build:

Pi02w fork (default)

Full build (override)

TOOL	INSTALLED	GITHUB	
SysMon	6.13.67.8-pi02w	6.13.67.9-pi02w	NEWER
44helper	0.0.174	0.0.175	NEWER
SysMon	–	6.13.74	other build

Quiet System

Restore

Figure 2.1 • Illustration. instmon's GitHub Updates section on a Pi Zero 2 W. The build choice appears only on a Pi Zero 2 W.

BUTTON	WHAT IT DOES
Check GitHub	Lists the newest copy of each tool on GitHub, and marks the ones newer than yours.
Update	Downloads and checks one file into instmon's library. Nothing is installed until you press Install on it.
Full Update	Updates every installed tool in one go, one tool at a time, with automatic roll-back if a new version fails.
Quiet System	Pauses SysMon, the Dashboard and 44helper during a big upload or install. The radio side keeps running, and everything comes back by itself after 30 minutes. Restore brings it back sooner.

TIP Before a Full Update, take a restore point in 44helper's Config tab. Full Update only replaces the tools, but a restore point means you can roll back any config change you make while trying out the new versions.

Updating Linux, ASL3 and DVSwitch packages is a separate job. 44helper's Update tab handles it (Chapter 10).

2.5 Removing the suite

```
# remove one tool
sudo python3 /usr/local/bin/sysmon.py --uninstall
# remove every tool except instmon
sudo bash uninstall_asl_dvs_all_v1_3.sh
```

Your settings in `/etc/asl_dvs/`, instmon's library and wifimon's config stay in place. Remove instmon last, with its own `--uninstall`.

UNDER THE HOOD

What install and update actually do

--install, per tool

- Copies the file into place, writes its `.service` unit, and writes the shared launcher if it is missing.
- Enables the service so it starts at boot, and starts it.
- `--uninstall` stops and removes the service and the program. It removes the launcher only once no other service uses it.

How the installer picks a build

1. `SYSMON_VARIANT` / `DASH_VARIANT` set to `pi02w` or `full`, if you set them;
2. otherwise, the build already installed;
3. otherwise, `pi02w` on a Pi Zero 2 W and `full` on any other Pi.

If no file of the chosen build is found, that tool is skipped with a warning; the other build is never used in its place. The installer looks in its own folder, the current folder, `/root`, `/home/pi`, your home folder, `/tmp` and instmon's library. `--non-interactive` (or `-y`) skips every question and takes the identity from `AUTO_CALLSIGN`, `AUTO_NODE` and `AUTO_LABEL`.

The checks on every download

Before instmon saves a file from GitHub, the file must match GitHub's file list. It must also have a version line and the right first line, pass a syntax check, and support `--install`.

Full Update, step by step

1. Downloads and checks every new file before anything changes.
2. Pauses the web tools.
3. For each tool in turn, saves a copy of the installed file in the library, runs the old `--uninstall`, and runs the new `--install`. If the new one fails, it puts the old one back and moves on.
4. Updates instmon last. A timer reinstalls the old instmon if the new one isn't answering after 90 seconds.
5. Ends with a summary. New install scripts go only to the Scripts library and are never run for you.

On a Pi Zero 2 W, the build choice decides which build Full Update installs. A tool installed in the other build is listed as **SWITCH** and replaced. On other Pis the choice is hidden and each tool keeps its build.

Settings that survive an update

Config files in `/etc` are never touched by an update. SysMon's Hardware tab options (HDMI and video driver off, lower GPU memory) are saved when SysMon is uninstalled during an update and put back on install, if the install follows within 30 minutes.